

# Gurleen Kaur

Vancouver, British Columbia | gurleenlakhman@gmail.com | gurleenkaur.dev | LinkedIn | GitHub  
Data Scientist (UBC Master of Data Science, 2026) with four years of applied ML and analytics experience across healthcare, clinical research, and neuroscience. Built a VR spatial navigation pipeline detecting early markers of cognitive decline at **84.5% balanced accuracy and 0.90 AUC** under leave-one-participant-out cross-validation, first-author manuscript in preparation. Holds three concurrent UBC research assistantships spanning time-series ML, digital pathology, and substance use epidemiology.

## Research Experience

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### NC4 Lab: Neural Circuits for Computation, Cognition and Control

Vancouver, BC

Research Assistant / Supervisor: Dr. Manu Madhav, UBC Biomedical Engineering

Apr 2026 – Present

- Built an ML pipeline on continuous 6DOF VR spatial navigation streams from 45 participants to detect early markers of cognitive decline, achieving 84.5% balanced accuracy and 0.90 AUC under leave-one-participant-out cross-validation with permutation testing ( $p = 0.001$ ); first-author manuscript in preparation.
- Benchmarked ROCKET time-series classifiers against XGBoost with handcrafted movement features (angular velocity, heading drift, rotation smoothness) and transformer-based sequence models, establishing the performance ceiling for automated screening on behavioural data.
- Ran feature ablation showing that a single spatial memory feature accounted for over 13 percentage points of balanced accuracy, isolating the strongest behavioural signal for clinical follow-up.
- Applied functional PCA (fPCA) and Dynamic Time Warping (DTW) for feature extraction on variable-length multivariate time series, reducing dimensionality and improving model interpretability for downstream clinical validation.

### Hirsch-Reinshagen Lab: Neuropathology & CNS Disorders

Vancouver, BC

Research Assistant / Supervisor: Dr. Veronica Hirsch-Reinshagen, UBC Pathology / VGH

Mar 2026 – Present

- Perform computational image analysis of multiplexed IHC slides from post-mortem human brain tissue using QuPath, quantifying protein co-expression and stain colocalization across brain regions to support neuropathological characterization of CNS disorders.
- Develop Python image analysis pipelines for automated spatial profiling of protein markers across brain regions, replacing manual annotation workflows and enabling scalable analysis across large post-mortem tissue cohorts.

### Addictions and Concurrent Disorders (ACD) Research Group

Vancouver, BC

Research Assistant / Supervisor: Dr. Michael Krausz, UBC Psychiatry

Nov 2025 – Present

- Perform data analysis and results comparison for a Delphi consensus study, synthesizing expert panel responses across rounds and producing the quantitative outputs used in the manuscript.
- Contribute under PI direction to a manuscript on opioid overdose and acquired brain injury (ABI), carrying out systematic review screening and structured data extraction across 50+ studies to synthesize evidence on overdose-related neurological injury.
- Analyzed spatial clustering and temporal patterns of fentanyl overdose events across vulnerable populations to identify high-risk windows and geographic hotspots, with findings intended to guide targeted harm reduction interventions.
- Contributing author on additional ACD manuscripts, including an early-stage paper on craving in substance use populations.
- Presented systematic review on stimulant co-use in opioid agonist therapy (OAT) at the PATHS To A Cure Neuroscience Research Conference, UBC; awarded Best Oral Presentation.

## Work Experience

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### Tandem Research / UBC MDS Industry Partner

Vancouver, BC

Data Science Capstone (INSPECT)

Apr 2026 – Jun 2026

- Built an end-to-end multimodal ML pipeline on GCP integrating CTPA imaging with EHR data for pulmonary embolism risk prediction, reaching 0.850 AUROC at 12 months and 0.809 AUROC at 6 months.
- Used GCS for data management, BigQuery as a centralized feature registry, and Vertex AI experiment tracking (AUROC, AUPRC) to make every result reproducible; delivered as a documented pipeline handoff to the partner team.
- Defined train/test strategy, cross-validation design, and evaluation metrics; produced model documentation covering assumptions, limitations, and failure modes for clinical and engineering stakeholders.

### Ronsare Research

Remote

Data Scientist

Jun 2022 – Jul 2025

- Engineered Python ETL pipelines processing large-scale healthcare claims data, reducing data preparation time by 40% and enabling reproducible downstream ML workflows across multiple client projects.
- Developed standardized preprocessing, validation, and profiling scripts for large-scale clinical datasets, cutting QA time and ensuring analytics-ready outputs meeting data governance requirements.
- Collaborated with data governance and IT teams to enforce metadata standards, reducing downstream data quality issues flagged during audits.

## KPMG India

*Data Scientist*

New Delhi, India (Hybrid)

Oct 2020 – May 2021

- Analyzed EMR/EHR datasets from hospital systems to surface patient outcome trends, producing stakeholder reports used directly by hospital management teams to support clinical decision-making.
- Built Python preprocessing pipelines for brain CT and MRI data and trained ML models for heart disease risk prediction; Random Forest classifier achieved 91.6% accuracy, supporting early-risk flagging for clinical triage.

## In-Med Prognostics

*Research and Development Intern*

Pune, India (Remote)

Mar 2021 – Jul 2021

- Built Python pipelines to preprocess and anatomically align brain CT images across 100 patients, standardizing imaging inputs and reducing manual preparation time ahead of downstream diagnostic model training.

## Selected Projects

### MolProp-UQ: Uncertainty-Aware Binding Affinity Prediction | *PyTorch Geometric, RDKit, Optuna*

- Trained a Graph Attention Network to predict EGFR binding affinity directly from molecular graphs constructed from ChEMBL bioactivity data, using Monte Carlo Dropout at inference to return calibrated uncertainty rather than single point estimates.
- Used Bemis-Murcko scaffold splitting so held-out molecules are structurally novel, testing generalization rather than memorization, and tuned hyperparameters with Bayesian optimization (Optuna TPE).

### BrainMRI-Net: Explainable Alzheimer's Classification | *PyTorch, Grad-CAM, OpenCV*

- Built a CNN classifying cognitively normal, MCI, and Alzheimer's patients from structural MRI, with subject-level train/test splitting to prevent leakage and macro F1 as the primary metric given class imbalance.
- Applied Grad-CAM so the model's attention patterns can be checked against known neuroanatomy, making the classifier auditable rather than accuracy-only.

## Technical Skills

**Languages:** Python, R, SQL, C, Bash

**ML & AI:** Supervised/Unsupervised Learning, XGBoost, Random Forests, Deep Learning, CNNs, Graph Neural Networks (GAT), Transformer Architectures, Time-Series Classification (ROCKET, DTW, fPCA), NLP, Retrieval-Augmented Generation (RAG), Clustering, Feature Engineering, Model Training & Evaluation, Explainability (Grad-CAM, feature ablation), Uncertainty Quantification (MC Dropout), Hyperparameter Optimization (Optuna)

**Statistics:** Statistical Modelling, Bayesian Inference, Regression Analysis, Experimental Design, Cross-Validation (LOPO, stratified k-fold), Permutation Testing, Geospatial and Spatiotemporal Analysis

**Clinical & Health Data:** Healthcare Claims (RWD), EHR/EMR Analysis, Medical Imaging (CT, MRI, CTPA), Clinical Data Preprocessing, Population Health, Systematic Review, Delphi Methods, Epidemiology, PHIPA/PIPEDA, TCPS 2

**Cloud & MLOps:** GCP (GCS, BigQuery, Vertex AI), Vertex AI Experiment Tracking, Docker, Git, GitHub, Linux/Unix, Reproducible Pipeline Design

**Libraries:** PyTorch, PyTorch Geometric, scikit-learn, pandas, NumPy, statsmodels, Matplotlib, Seaborn, ggplot2, OpenCV, PIL, tslearn, skfda, RDKit, Optuna

**Digital Pathology:** QuPath, Fiji/ImageJ, Immunohistochemistry (IHC) Image Analysis, Stain Colocalization, Spatial Profiling

**Databases:** PostgreSQL, MySQL, MongoDB    **Visualization & BI:** Tableau, Power BI, ggplot2, Matplotlib

**Tools:** Jupyter, RStudio, pgAdmin, LaTeX, Git, GitHub, Docker, Linux, Overleaf

## Education

### University of British Columbia

*Master of Data Science*

Vancouver, BC

Aug 2025 – Jun 2026

Coursework: Statistical Inference, Regression Analysis, Experimental Design, Bayesian Methods, Supervised & Unsupervised Learning, Spatial & Temporal Data, Data Visualization, Databases, Natural Language Processing.

### RRC Polytech

*Advanced Diploma in Technology Management*

Winnipeg, MB

Jan 2023 – Aug 2023

### MIT Pune

*BTech, Electronics and Telecommunication (AI Focus)*

Pune, India

Jul 2018 – Jul 2022

## Certifications & Awards

**Best Oral Presentation**, PATHS To A Cure Neuroscience Research Conference, UBC (2026)

**TCPS 2: CORE 2022**, Ethical Conduct for Research Involving Humans, Government of Canada (Mar 2026)

**CFA Program** (in progress) | **Google Data Analytics Professional Certificate**, Google | **Data Science: R Basics, Machine Learning, Data Visualization**, HarvardX